

SAN ANTONIO INTL, TX, USA

WMO: 722530

Lat: 29.544N

Lon: 98.484W

Elev: 789

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.0	33.1	11.2	10.0	42.4	15.7	12.5	47.1	24.2	57.7	20.8	55.0	7.7	10	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.7	99.3	73.6	97.7	73.6	95.9	73.8	78.1	88.0	77.4	87.2	76.9	86.4	10.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	139.5	80.2	75.2	136.1	80.0	74.5	132.8	80.0	42.0	88.2	41.3	87.2	40.7	86.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	19.0	17.4	DB	24.1	103.0	3.3	2.7	21.7	104.9	19.8	106.4	17.9	107.9	15.5	109.9
			WB	21.3	79.6	3.0	1.0	19.1	80.3	17.4	80.9	15.7	81.4	13.5	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.3	52.9	57.1	63.1	70.1	77.3	82.9	85.0	85.8	80.4	71.7	61.7	54.1	(6)
(7)		DBStd	13.65	8.99	9.84	8.52	6.84	5.42	3.63	3.06	3.28	5.17	7.66	8.82	8.93	(7)
(8)		HDD50	196	73	40	11	1	0	0	0	0	0	1	13	57	(8)
(9)		HDD65	1352	384	252	137	30	2	0	0	0	1	30	162	354	(9)
(10)		CDD50	7590	163	240	418	603	846	988	1087	1111	911	673	365	185	(10)
(11)		CDD65	3270	9	31	79	183	383	538	622	646	462	237	64	16	(11)
(12)		CDH74	34561	57	209	485	1385	3463	6065	7509	8099	4846	2000	369	74	(12)
(13)		CDH80	16070	4	46	100	425	1398	2923	3887	4352	2239	640	52	4	(13)
(14)	Wind	WSAvg	8.5	7.7	8.3	9.2	9.8	10.0	9.6	9.0	8.3	7.1	7.6	7.6	7.3	(14)
(15)	Precipitation	PrecAvg	31.8	1.7	1.9	1.8	2.4	4.1	3.9	2.2	2.6	3.3	3.7	2.6	1.8	(15)
(16)		PrecMax	52.3	8.5	6.4	6.1	8.8	12.8	12.0	16.9	11.1	13.1	18.1	9.4	14.0	(16)
(17)		PrecMin	16.4	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	8.9	1.7	1.5	1.3	1.9	3.1	3.0	3.3	2.5	2.8	3.5	2.2	2.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	85.4	86.7	93.0	96.4	100.4	100.9	101.6	99.8	92.2	84.8	80.0	(19)
(20)			MCWB	60.2	61.1	65.3	69.1	73.4	73.8	72.7	73.3	72.0	71.9	69.7	64.8	(20)
(21)		2%	DB	75.0	79.7	82.6	88.1	93.3	97.2	98.8	99.4	96.1	89.1	81.2	75.9	(21)
(22)			MCWB	59.8	62.2	65.1	68.5	73.0	73.9	73.4	73.9	72.9	71.9	67.4	63.2	(22)
(23)		5%	DB	71.5	75.8	79.4	84.6	90.4	94.8	97.0	97.8	93.4	86.6	78.3	72.7	(23)
(24)			MCWB	59.9	62.1	64.8	68.3	72.9	74.1	73.8	73.8	72.9	70.9	66.4	62.7	(24)
(25)		10%	DB	68.3	71.9	76.0	81.8	87.6	92.3	94.8	95.9	90.7	83.7	75.4	69.4	(25)
(26)			MCWB	59.2	61.2	63.8	67.6	72.5	74.2	74.2	74.0	72.7	69.7	65.4	62.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	69.3	71.2	75.3	78.2	78.8	78.5	78.4	78.1	77.2	73.5	70.2	(27)
(28)			MCD B	71.5	75.1	78.8	84.2	88.5	89.2	88.2	89.4	86.4	83.8	79.2	74.9	(28)
(29)		2%	WB	66.4	67.9	69.6	73.3	76.8	77.8	77.7	77.5	77.1	75.9	71.8	68.3	(29)
(30)			MCD B	70.4	73.2	75.7	81.5	86.5	88.2	87.7	87.9	85.1	82.2	76.9	72.2	(30)
(31)		5%	WB	64.2	66.4	68.5	72.1	75.5	77.1	77.1	77.0	76.4	74.9	70.1	66.4	(31)
(32)			MCD B	68.6	71.4	74.4	79.4	84.2	87.1	87.0	87.2	84.3	81.0	75.2	70.3	(32)
(33)		10%	WB	61.5	64.7	67.2	70.9	74.7	76.3	76.4	76.4	75.6	73.6	68.3	63.6	(33)
(34)			MCD B	66.4	69.7	72.9	77.6	83.0	85.7	85.9	86.4	83.5	79.5	73.6	67.9	(34)
(35)	Mean Daily Temperature Range		MDBR	21.2	20.9	20.5	20.5	18.8	18.5	19.0	19.7	19.3	20.0	20.1	20.7	(35)
(36)		5% DB	MCD BR	25.6	25.0	23.1	23.1	21.5	21.4	21.8	21.9	21.9	20.1	21.1	24.0	(36)
(37)			MCWBR	14.9	13.4	11.2	10.2	6.9	5.0	4.6	4.7	5.7	7.8	11.1	14.3	(37)
(38)		5% WB	MCD BR	18.3	18.2	17.9	17.5	17.3	17.2	17.9	19.0	17.3	15.8	16.6	18.1	(38)
(39)			MCWBR	13.6	12.3	10.6	9.2	7.1	5.7	4.8	4.9	5.7	7.7	10.7	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.332	0.356	0.399	0.444	0.428	0.436	0.421	0.414	0.373	0.360	0.318	(40)	
(41)		taud	2.502	2.487	2.433	2.330	2.249	2.346	2.360	2.382	2.355	2.458	2.427	2.536	(41)	
(42)		Ebn at Noon	290	297	295	284	270	273	271	274	273	279	274	287	(42)	
(43)		Edh at Noon	29	31	36	41	44	40	39	38	38	32	31	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	959	1126	1423	1692	1824	2094	2110	2017	1653	1367	1028	851	(44)	
(45)		RadStd	107	201	144	169	189	138	184	116	166	182	116	108	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+112	(46)
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	(47)

Nomenclature: See separate page